Benchmarking Variational Quantum Circuits for Probability Distribution Loading

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Abstract

Background: Quantum state preparation, the encoding of classical probability distributions into quantum amplitude registers, is a foundational subroutine for quantum Monte Carlo, quantum finance, and quantum machine learning. While variational quantum circuits (VQCs) have emerged as the dominant NISQ-era approach, no prior study has systematically quantified how VQC loading performance varies across distribution shapes, circuit architectures, and optimizers.

Purpose: This study presents a comprehensive benchmarking to establish an empirical difficulty ordering for VQC-based probability distribution loading and to provide actionable circuit design guidance for quantum practitioners.

Methods: We evaluate three hardware-efficient Ry-CZ ansatz topologies (linear, circular, alternating) across six target distributions spanning the complete shape spectrum: symmetric (uniform, binomial), asymmetric (Poisson $\boldsymbol{\lambda = 1.5}$, $\boldsymbol{\lambda = 2.5}$, geometric), and multimodal (bimodal Gaussian mixture). Using Qiskit 1.x simulators, we conduct 2,880 experimental runs (6 distributions $\times$ 3 architectures $\times$ 4 depths $\times$ 2 qubit counts $\times$ 2 optimizers $\times$ 10 random seeds), measuring fidelity, convergence rates, and resource requirements.

Results: Distribution shape is the primary determinant of loading difficulty. Uniform achieves the highest mean fidelity ($\mathbf{0.8617 \pm 0.0607}$), followed by Geometric ($\mathbf{0.7979 \pm 0.1523}$), Poisson($\boldsymbol{\lambda = 2.5}$) ($\mathbf{0.7715 \pm 0.1379}$), Bimodal ($\mathbf{0.7593 \pm 0.1646}$), Poisson($\boldsymbol{\lambda = 1.5}$) ($\mathbf{0.7542 \pm 0.1480}$), and Binomial ($\mathbf{0.6981 \pm 0.1433}$). Under best-practice conditions (COBYLA optimizer at depth 4), performance improves substantially: Geometric reaches $\mathbf{0.9077 \pm 0.0345}$, Uniform $\mathbf{0.9015 \pm 0.0380}$, and Bimodal $\mathbf{0.8577 \pm 0.0428}$. Architecture differences are minimal (maximum fidelity gap $\boldsymbol{\Delta F = 0.031}$). COBYLA substantially outperforms SPSA ($\mathbf{0.8482 \pm 0.0739}$ versus $\mathbf{0.6994 \pm 0.1619}$), converging in $\mathbf{52 \pm 15}$ iterations versus SPSA's unconverged 500-iteration budget. Fidelity degrades sharply from $\boldsymbol{n = 3}$ to $\boldsymbol{n = 4}$ qubits (drops of $\mathbf{-0.085}$ to $\mathbf{-0.206}$).

Implications and Impact: These findings provide empirical difficulty ordering for VQC-based distribution loading. For practitioners: use COBYLA, prefer alternating connectivity for asymmetric distributions, match depth to distribution entropy, and anticipate significant re-optimization overhead when scaling qubits. The results have immediate implications for quantum generative modeling and Monte Carlo applications where asymmetric and multimodal distributions predominate.

Keywords: Variational quantum algorithms, Parameterized quantum circuits, Quantum state preparation, Probability distribution loading, Qiskit

1 Introduction

For fundamentally probabilistic problems, quantum computing offers exponential or polynomial speedups over classical algorithms. Major quantum computing operations, including quantum Monte Carlo sampling, quantum Bayesian inference, quantum generative modeling, and option pricing algorithms, all share a common requirement: efficiently encoding a classical probability distribution (e.g., Poisson or Binomial) into a quantum register (Sano & Hamamura, 2025). This process is not a pre-processing step but a fundamental one; it often constitutes the computational bottleneck that determines whether a quantum algorithm will be more efficient than its classical counterpart (Zoufal, Lucchi, & Woerner, 2019). In the literature, this process is variously referred to as *quantum distribution loading*, *quantum state preparation*, or *amplitude encoding*.

For this task specifically, Parameterized Quantum Circuits (PQCs), also known as variational quantum circuits, have prevailed due to their compatibility with Noisy Intermediate-Scale Quantum (NISQ) devices and inherent flexibility (Barthe, Grossi, Vallecorsa, Tura, & Dunjko, 2025). PQCs can generate both asymmetric and symmetric distributions, and variational solvers can optimize circuit parameters to closely match target distributions (Dasgupta & Paine, 2022). VQCs for state preparation and generative modeling have been discussed considerably in recent years (Barthe et al., 2025; Dasgupta & Paine, 2022; Raj, Durge, Mallapur, Kumar, & Raina, 2026).

The architecture by Dasgupta and Paine (2022) provides the main circuit and methodological framework followed here. They propose parameterized quantum circuits that directly generate both symmetric and asymmetric probability distributions by tracing the trajectory of quantum states under successive single- and two-qubit gates, and then optimizing the gate parameters with a classical variational solver (COBYLA). For symmetric distributions (e.g., Gaussian-like), their construction uses a single layer of RY rotations followed by carefully placed CNOTs exploiting symmetry to avoid circuit elements that scale exponentially with qubit number, yielding an adaptive variational circuit that needs only n RY gates and $n-1$ CNOTs for n qubits. Asymmetric (skewed) distributions such as log-normal or chi-square are handled by conditioning rotations and swaps on the most significant qubits, decomposing the distribution into regions (e.g., tail versus peak side) and adding controlled RY layers

for fine-tuning skewness. Throughout, a hybrid quantum-classical loop updates gate angles to minimize distance between the generated and target distributions, and the authors benchmark this Adaptive VC against a standard Ry-CZ variational ansatz, showing improved or comparable performance on Gaussian, log-normal, and chi-square targets in both simulation and on IBM hardware. Consequently, this architecture serves as the primary template in this work for both circuit design (gate layout, symmetry exploitation, scaling) and algorithmic methodology (variational training and distribution matching).

Despite growing interest in VQC-based distribution loading, the literature lacks a systematic, side-by-side comparison of how variational circuits perform across different distribution types, circuit architectures, and optimization strategies. Existing studies typically evaluate a single circuit design on one or two distribution types, making it impossible to draw conclusions about which distributions are inherently harder to load, whether circuit connectivity patterns affect loading quality, or which optimizer should be preferred for this specific task. Reviews have also underlined the importance of benchmarking circuit resources, as it has been shown that increasing circuit depth and gate count may lead to better proximity to desired distributions but at the expense of more noise and optimization difficulties (Raj et al., 2026; Sano & Hamamura, 2025). Optimization efficiency is especially important as it describes the performance of classical optimizers (e.g., COBYLA and SPSA) in the context of variational quantum algorithms. Additionally, hardware experimentation shows how error accumulation, limited qubit connectivity, noise, and general interference greatly reduce the quality of generated distributions (Dasgupta & Paine, 2022; Sano & Hamamura, 2025).

This study is designed to fill precisely these gaps through implementing and benchmarking the variational circuit designs illustrated by Dasgupta and Paine (2022) in Qiskit. Specifically, the study is interested in comparing and contrasting different generation techniques of basic symmetric distributions (such as binomial distributions) and asymmetric distributions (such as Poisson or geometric distributions). Additionally, it conducts a depth-fidelity scaling analysis across all six distribution types, characterizing saturation behavior and identifying minimum depth requirements per distribution. Finally, all implementations are fully reproducible in Qiskit 1.x with open-source code, providing a benchmarking framework the community can extend.

The remainder of this paper is organized as follows. Section 2 provides the theoretical framework, including the mathematical foundations of amplitude encoding, variational circuits, and the evaluation metrics used. Section 3 describes the simulation design and experimental methodology in detail. Section 4 presents the algorithmic formulation and pseudocode for each component of the framework. Section 5 presents results. Section 6 discusses implications, limitations, and future directions. Section 7 concludes.

2 Theoretical Framework

Before describing what we actually build and measure, it is worth being precise about what quantum distribution loading means mathematically, and why the variational approach is the natural one to take in the NISQ era. This section develops the necessary

formalism in the order it is needed: first the state-space encoding that makes the whole enterprise possible, then the parametric circuit family we optimize over, then the three connectivity topologies we compare, then the optimization objective and the two optimizers employed, and finally the five metrics used to judge success.

2.1 Quantum State Representation and Distribution Loading

An n-qubit quantum system lives in a 2^n-dimensional Hilbert space. Any pure state can be written as

$$|\psi\rangle = \sum_{i=0}^{2^n-1} \alpha_i \, |i\rangle, \qquad \sum_i |\alpha_i|^2 = 1, \tag{1}$$

where $|i\rangle$ denotes the i-th computational basis vector. Measuring Equation (1) in the computational basis returns outcome i with probability $P(i) = |\alpha_i|^2$. This measurement rule makes the encoding strategy immediate: given a normalized target distribution $P_{\text{des}} = (p_0, \ldots, p_{N-1})$ with $N = 2^n$, we seek a quantum state whose measurement statistics reproduce P_{des}, that is,

$$|\psi_{\text{des}}\rangle = \sum_{i=0}^{N-1} \sqrt{p_i} \, |i\rangle. \tag{2}$$

This is amplitude encoding (Grover & Rudolph, 2002). Exact preparation of an arbitrary state of the form Equation (2) in general requires $\mathcal{O}(2^n)$ gates, which immediately motivates the approximate variational approach: rather than constructing the target state exactly, we search over a parametrically restricted family of circuits for the member that best approximates it, at a gate cost that scales polynomially rather than exponentially with n.

2.2 Parameterized Quantum Circuits

Parameterized quantum circuits act as flexible function approximators over quantum amplitudes (Benedetti, Lloyd, Sack, & Fiorentini, 2019). Their expressive capacity, how rich a set of probability distributions they can represent, depends jointly on depth, gate set, and entanglement structure (Sim, Johnson, & Aspuru-Guzik, 2019). We study an n-qubit, L-layer ansatz whose full unitary is

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} \Big[U_{\text{ent}}(\mathcal{E}) \bigotimes_{j=0}^{n-1} R_Y\big(\theta_j^{(\ell)}\big) \Big] \cdot \bigotimes_{j=0}^{n-1} R_Y\big(\theta_j^{(0)}\big), \tag{3}$$

where $R_Y(\theta) = e^{-i\theta\sigma_Y/2}$ is a single-qubit rotation about the Y-axis and $U_{\text{ent}}(\mathcal{E})$ is an entanglement block whose structure depends on the connectivity graph $\mathcal{E}$, described in the next subsection. The circuit places an initial rotation layer before any entanglement, then interleaves rotations and entanglers for each of the L subsequent layers. The total trainable parameter count is therefore

$$|\boldsymbol{\theta}| = n(L+1), \tag{4}$$

which grows only linearly in both n and L, a favourable property for optimization. Applying $U(\boldsymbol{\theta})$ to the all-zero state and measuring produces the generated distribution

$$P_{\text{gen}}(i;\boldsymbol{\theta}) = \left|\langle i \mid U(\boldsymbol{\theta}) \mid 0\rangle^{\otimes n}\right|^2, \tag{5}$$

which the optimization loop drives toward P_{des}.

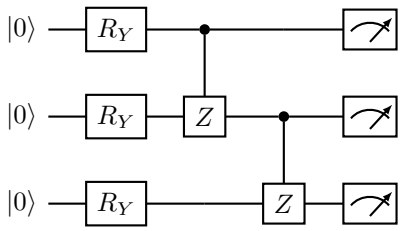
2.3 Entanglement Architectures

The connectivity graph $\mathcal{E}$ in Equation (3) determines which pairs of qubits interact in each entanglement block, and therefore how quickly correlations can be built up across the distribution's outcome space. We compare three topologies, illustrated for $n = 3$ qubits and $L = 1$ layer. All three share the same gate set and parameter count Equation (4), so connectivity is isolated as the sole architectural variable in the comparisons reported in Section 5.

Linear (Nearest-Neighbour)

$$\mathcal{E}_{\text{lin}} = \{(j,\, j+1) \mid 0 \le j < n-1\}, \tag{6}$$

placing $n-1$ CZ gates per layer at $\mathcal{O}(n)$ cost. This mirrors the coupling maps of IBM superconducting hardware, making the linear architecture the most hardware-realistic of the three.

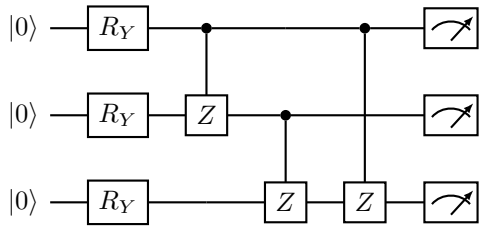


Circular (Ring)

Adding a single CZ gate between the last and first qubit closes the chain into a ring,

$$\mathcal{E}_{\text{circ}} = \{(j,\, (j+1) \bmod n) \mid 0 \le j < n\}, \tag{7}$$

giving n entanglers per layer. This periodic boundary introduces an additional correlation pathway between qubits 0 and $n-1$ without changing the parameter count Equation (4).

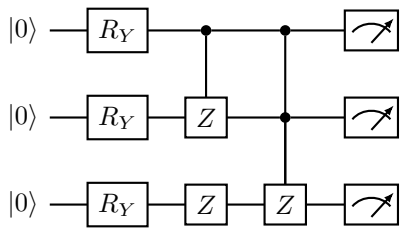


Alternating (Brickwork)

The alternating, or brickwork, topology uses a shifting pattern: even layers apply CZ to $(q_0, q_1), (q_2, q_3), \ldots$ and odd layers to $(q_1, q_2), (q_3, q_4), \ldots$, so every pair of adjacent qubits interacts in alternate layers. This corresponds to the "all-to-all" connectivity in the mathematical formalism,

$$\mathcal{E}_{\text{alt}} = \{(j, k) \mid 0 \leq j < k < n\}, \tag{8}$$

and has been shown to maximize expressibility for fixed depth (Sim et al., 2019). For $n = 3$ this places three CZ gates per layer at $\mathcal{O}(n^2)$ cost, increased entanglement density at the expense of deeper circuits.



2.4 Variational Objective and Optimization

Having defined how the circuit generates a distribution via Equation (5), the optimization problem is to minimize a measure of discrepancy between P_{gen} and P_{des}. We use the squared Euclidean (L_2) distance,

$$\mathcal{L}(\boldsymbol{\theta}) = \|P_{\text{gen}}(\boldsymbol{\theta}) - P_{\text{des}}\|_2^2, \tag{9}$$

which is differentiable with respect to $\boldsymbol{\theta}$ and treats all outcome bins symmetrically. Because P_{gen} is obtained from finite-shot measurements, the objective Equation (9) is inherently stochastic, which influences the choice of optimizer.

COBYLA.

Constrained Optimization BY Linear Approximation is a gradient-free method that builds successive linear models of the objective within a trust region. It requires only function evaluations, never gradient information, making it well suited to noisy, discrete quantum outputs (Cerezo et al., 2021).

SPSA.

Simultaneous Perturbation Stochastic Approximation estimates the full gradient vector from exactly two function evaluations per iteration, regardless of parameter dimension. At iteration k, a Rademacher perturbation vector $\boldsymbol{\Delta}_k \in \{-1, +1\}^p$ is drawn at random, and the gradient estimate is

$$\hat{g}_k = \frac{\mathcal{L}(\boldsymbol{\theta}_k + c_k \boldsymbol{\Delta}_k) - \mathcal{L}(\boldsymbol{\theta}_k - c_k \boldsymbol{\Delta}_k)}{2 c_k \boldsymbol{\Delta}_k}, \tag{10}$$

with the parameter update $\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - a_k \hat{g}_k$. The gain sequences follow the theoretically optimal schedule from Spall (1998),

$$a_k = \frac{a}{(A + k + 1)^{\alpha}}, \qquad c_k = \frac{c}{(k + 1)^{\gamma}}, \tag{11}$$

with $\alpha = 0.602$, $\gamma = 0.101$. Tracking $\mathrm{Var}[\hat{g}_k]$ across training serves as a barren-plateau diagnostic: exponentially vanishing gradient variance as n increases is the hallmark of trainability breakdown in deep circuits (Cerezo et al., 2021).

2.5 Evaluation Metrics

Five complementary metrics quantify how well P_{gen} approximates P_{des} over $N = 2^n$ outcomes. Fidelity F is the primary overlap measure,

$$F = \left(\sum_{k=0}^{N-1} \sqrt{P_{\text{gen}}(k)\, P_{\text{des}}(k)} \right)^2, \tag{12}$$

ranging from 0 (no overlap) to 1 (perfect match). The optimization objective $\mathcal{L}$ Equation (9) is reported directly as a distance measure. Information-theoretic divergences provide complementary views: the asymmetric Kullback-Leibler divergence

$$D_{\text{KL}} = \sum_k P_{\text{des}}(k) \ln \frac{P_{\text{des}}(k)}{P_{\text{gen}}(k)}, \tag{13}$$

penalizes outcomes where P_{gen} underestimates the target, while the symmetric Jensen-Shannon divergence

$$D_{\text{JS}} = \tfrac{1}{2} D_{\text{KL}}(P_{\text{des}} \,\|\, M) + \tfrac{1}{2} D_{\text{KL}}(P_{\text{gen}} \,\|\, M), \quad M = \tfrac{1}{2}(P_{\text{des}} + P_{\text{gen}}), \tag{14}$$

is bounded in $[0, 1]$ and numerically stable near-zero probability bins. Finally, total variation

$$D_{\text{TV}} = \tfrac{1}{2} \| P_{\text{gen}} - P_{\text{des}} \|_1 \tag{15}$$

measures the maximum probability that can be assigned differently by the two distributions and has a direct operational interpretation in binary discrimination. Together,

F, D_{KL}, D_{JS}, and D_{TV} capture complementary aspects of distributional discrepancy, and reporting all four guards against any single metric that could look good for the wrong reasons.

3 Simulation Design

The theoretical framework of Section 2 sets up the optimization problem; this section specifies exactly how that problem is instantiated, executed, and measured. Every design choice below is made explicit so that the full experiment can be reproduced from scratch.

3.1 Software and Hardware Environment

All experiments are conducted in simulation using Qiskit 1.2.4 and Qiskit-Aer. We use the shot-based simulator throughout because the objective function Equation (9) is evaluated from measurement counts in exactly the same way it would be on real hardware; sampling noise is therefore part of the experimental signal rather than an artifact to be suppressed. Real quantum hardware is deliberately excluded for three reasons. First, device-specific noise would confound algorithmic comparisons, making it impossible to attribute performance differences to architecture or optimizer rather than gate fidelity or connectivity constraints. Second, fixed random seeds make simulated results exactly reproducible. Third, the scale of the experiment, 2,880 individual runs, is impractical on any current cloud quantum platform.

Qiskit 1.x replaced the deprecated `execute()` function with the `Sampler` primitive, and all circuits in this work use `qiskit.primitives.Sampler` accordingly. Each circuit is executed with `shots` = 1,000, giving a per-bin sampling standard deviation of approximately 3% while keeping wall-clock cost manageable. The scientific stack comprises NumPy 1.26, SciPy 1.12, and Matplotlib 3.8.

3.2 Target Distributions

Six target distributions are chosen to span a broad range of shapes, symmetry classes, and tail behaviours. Each is discretized to $N = 2^{n_{\text{qubits}}}$ outcomes (8 bins for $n = 3$, 16 for $n = 4$) and normalized to sum to 1. Laplace smoothing with $\varepsilon = 10^{-8}$ is applied before computing divergences to prevent $\log(0)$ in Equations (13) and (14). Figure 1 shows the theoretical shapes of all six target distributions.

Binomial$(n_{\text{qubits}}, 0.5)$ is a symmetric, unimodal distribution computed via `scipy.stats.binom.pmf` and zero-padded to N outcomes. Its near-Gaussian bell shape is close to the natural output of a shallow Ry layer.

Uniform places equal mass $1/N$ on every outcome. While trivially the distribution of an n-qubit Hadamard state, generating it with a Ry-CZ ansatz requires the optimizer to suppress the structured non-uniformity produced by Ry rotations from $|0\rangle^{\otimes n}$.

Poisson$(\lambda = 1.5)$ and **Poisson**$(\lambda = 2.5)$ are both right-skewed, computed via `scipy.stats.poisson.pmf` and truncated to N outcomes; discarded tail mass is less

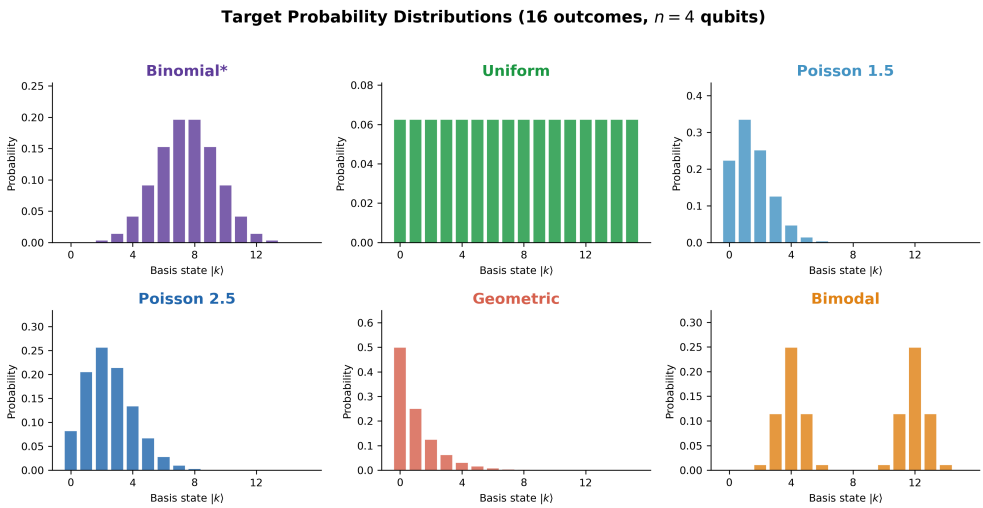


Fig. 1 Theoretical probability distributions used as loading targets. Each distribution is discretized to $N = 2^n$ outcomes for $n = 3$ (8 bins) and $n = 4$ (16 bins). The distributions span symmetric (Binomial, Uniform), asymmetric (Poisson with $\lambda = 1.5$ and $\lambda = 2.5$, Geometric), and multimodal (Bimodal Gaussian mixture) shape classes.

than 0.001 for both. Including two Poisson variants isolates the effect of skewness magnitude independently of distribution family.

Geometric($p = 0.5$), defined by $P(k) = (1-p)^{k-1}p$ for $k = 1, 2, \ldots$ and truncated to N outcomes, has a monotonically decreasing profile that forces the circuit to concentrate probability entirely in the low-index bins.

Bimodal (Gaussian mixture) is constructed as

$$P(k) \propto \exp\left(-\frac{(k-\mu_1)^2}{2\sigma^2}\right) + \exp\left(-\frac{(k-\mu_2)^2}{2\sigma^2}\right), \tag{16}$$

with $\mu_1 = \lfloor N/4 \rfloor$, $\mu_2 = \lfloor 3N/4 \rfloor$, $\sigma = 0.8$. Two separated probability peaks require simultaneous constructive interference at two distant regions of the outcome space, the most demanding encoding task, and the one whose difficulty ordering we expect to be largest.

3.3 Experimental Design

The study uses a fully crossed factorial design over six factors, summarized in Table 1. Every combination of levels is run ten times, once per random seed from 0 to 9. Replication across seeds quantifies initialization sensitivity and allows reliable mean-and-standard-deviation reporting. The three circuit architectures share the same gate set and parameter count Equation (4), isolating connectivity as the sole architectural

variable; depth is varied from 1 to 4 to characterize fidelity-depth trade-offs and saturation points; the two qubit counts test whether observations hold as the Hilbert space dimension doubles from 8 to 16 bins.

Table 1 Full factorial experimental design. The product of all level counts gives the total run count.

Factor	Levels	Count
Distribution	Binomial, Uniform, Poisson(1.5), Poisson(2.5), Geometric, Bimodal	6
Architecture	Linear, Circular, Alternating	3
Depth D	1, 2, 3, 4	4
n_{qubits}	3, 4	2
Optimizer	COBYLA, SPSA	2
Random seed	0–9	10
Total		$6 \times 3 \times 4 \times 2 \times 2 \times 10 =$ **2,880**

3.4 Optimization Protocol

Both optimizers minimize the cost function Equation (9) built from 1,000-shot evaluations of the current parameter vector. Initial parameters are drawn uniformly from $[0, 2\pi]$ using the run seed. COBYLA runs with `maxiter` $= 500$, `rhobeg` $= 1.0$, and `tol` $= 10^{-4}$. SPSA uses the gain schedule Equation (11) with $a = 0.1$, $c = 0.1$, $A = 10$, and `maxiter` $= 500$; its gradient-variance trace is logged for barren-plateau analysis. In both cases the full cost history is recorded. A run is declared converged when the final cost satisfies $\mathcal{L} < 0.15$; non-converged runs are retained in all statistics but flagged separately.

3.5 Evaluation Protocol and Workflow

Figure 2 shows the complete workflow for a single experimental run. After selecting a configuration from Table 1 and fixing the random seed, initial parameters are drawn from $\mathcal{U}[0, 2\pi]$ and the optimization loop begins. At each iteration the circuit is built according to Equation (3), executed for 1,000 shots, and the resulting counts converted to P_{gen} via Equation (5); the cost Equation (9) is then passed to the optimizer. Upon termination the best parameter vector is used for a final clean 1,000-shot evaluation, and all five metrics Equations (12)–(15) are written to a JSON record for later aggregation.

Beyond fidelity and divergences, each run records the R_Y gate count, CZ gate count, total gate count, post-transpilation depth, and parameter count. These resource statistics allow fidelity gains to be placed in the context of circuit cost, which matters practically whenever results are intended for eventual deployment on noisy hardware.

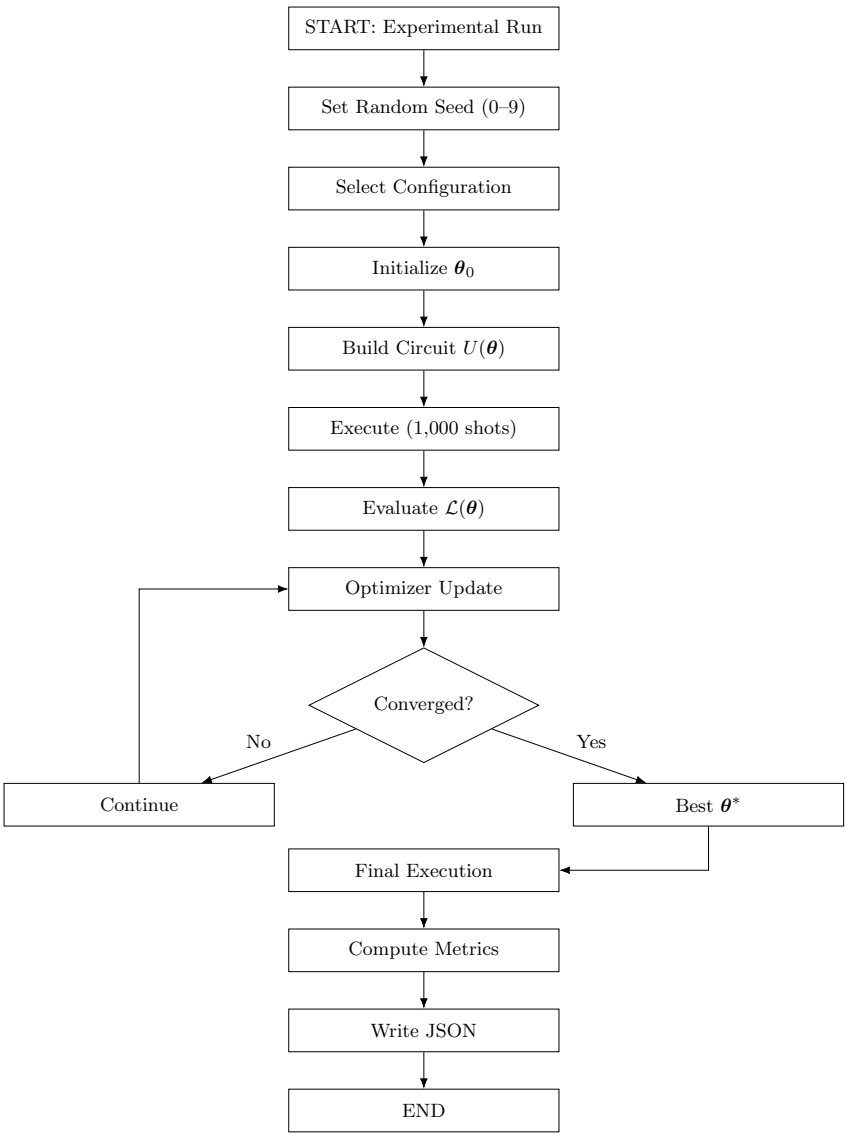


Fig. 2 Experimental workflow for a single configuration and seed. The inner loop (Build → Execute → Cost → Optimizer Update) repeats until $\mathcal{L} < 0.15$ or 500 iterations are exhausted. Upon termination, the best $\boldsymbol{\theta}^*$ drives a clean final evaluation before all metrics are logged.

4 Algorithmic Formulation

This section presents the five algorithms that together implement the benchmarking framework described in Section 3. Each algorithm is given in pseudocode form and derives directly from the mathematical specification of Section 2. Together they constitute a self-contained, reproducible specification of the experimental pipeline.

4.1 Target Distribution Generation

Algorithm 1 formalizes the discretization and normalization procedure for all six target distributions. The output is a normalized probability vector $P_{\text{des}} \in \mathbb{R}^{2^n}$ satisfying $\sum_k P_{\text{des}}(k) = 1$ and $P_{\text{des}}(k) \geq \varepsilon$ (the Laplace floor prevents log(0) in Equations (13) and (14) during evaluation).

Algorithm 1 Generate Target Distribution

Require: Distribution name d, qubit count n, smoothing $\varepsilon = 10^{-8}$
Ensure: Normalized vector $P_{\text{des}} \in \mathbb{R}^{2^n}$

```
N ← 2^n;   x ← [0, 1, ..., N − 1]
if d = binomial then
    P ← BinomPMF(x; n, p=0.5);   zero-pad to length N
else if d = uniform then
    P(k) ← 1/N for all k
else if d = poisson then
    P ← PoissonPMF(x; λ);   add tail mass to P(N−1)
else if d = geometric then
    P(k) ← (1 − p)^(k−1) p;   truncate and renormalize to N outcomes
else if d = bimodal then
    μ1 ← ⌊N/4⌋;  μ2 ← ⌊3N/4⌋;  σ ← 0.8
    for k = 0, ..., N − 1 do
        P(k) ← exp(−(k−μ1)^2/(2σ^2)) + exp(−(k−μ2)^2/(2σ^2))   ▷ Eq. (16)
    end for
end if
P ← max(P, ε)   ▷ Laplace smoothing
return P / ‖P‖_1   ▷ Normalize to sum to 1
```

4.2 Circuit Construction

Algorithm 2 implements the unitary $U(\boldsymbol{\theta})$ of Equation (3). It accepts a parameter vector of length $n(L+1)$ Equation (4), builds an initial Ry layer, then repeats L blocks of entanglement followed by Ry rotations. The entanglement block $U_{\text{ent}}(\mathcal{E})$ is dispatched according to the connectivity choice, implementing any of the three topologies Equations (6)–(8).

4.3 Cost Function Evaluation

Algorithm 3 evaluates the optimization objective Equation (9) for a given parameter vector. It invokes Algorithm 2 to construct the circuit, executes it for $S = 1{,}000$ shots via the Qiskit `Sampler` primitive, and converts measurement counts to a probability vector from which the L_2 cost is computed. This function is the oracle called by both COBYLA and SPSA at each iteration.

Algorithm 2 Build Ry-CZ Circuit

Require: Parameters $\boldsymbol{\theta} \in \mathbb{R}^{n(L+1)}$, qubits n, depth L, topology $\mathcal{E}$
Ensure: Quantum circuit $\mathcal{C}$ implementing $U(\boldsymbol{\theta})$ Equation (3)

```
C ← QuantumCircuit(n)
idx ← 0
for j = 0, ..., n − 1 do                    ▷ Initial Ry layer (layer ℓ = 0)
    C.ry(θ_idx, j);   idx += 1
end for
for ℓ = 1, ..., L do                        ▷ Entanglement block
    for all (j, k) ∈ E do
        C.cz(j, k)
    end for
    for j = 0, ..., n − 1 do                ▷ Ry layer for this depth
        C.ry(θ_idx, j);   idx += 1
    end for
end for
return C
```

Algorithm 3 Evaluate L_2 Cost

Require: Parameters $\boldsymbol{\theta}$, target P_{des}, shots $S = 1{,}000$, topology $\mathcal{E}$, depth L
Ensure: Scalar cost $\mathcal{L}(\boldsymbol{\theta})$ Equation (9)

```
C ← BuildCircuit(θ, n, L, E)                ▷ Algorithm 2
Add measurement gates to all qubits of C
counts ← Sampler.run(C, shots=S)
P_gen(i) ← counts[i] / S for each outcome i
return ||P_gen − P_des||_2^2
```

4.4 SPSA Optimization

Algorithm 4 formalizes the SPSA optimization loop, implementing the gradient estimate Equation (10) and the gain schedules Equation (11). Beyond driving the parameter update, the algorithm records $\mathrm{Var}[\hat{g}_k]$ at every iteration. This variance trace is used in the barren-plateau analysis of Section 5.8: a monotone decay toward zero is the signature of a flat optimization landscape, while maintained variance indicates healthy gradient signal.

4.5 Full Factorial Runner

Algorithm 5 coordinates the complete 2,880-run experiment. For each element of the Cartesian product of factor levels (Table 1) it fixes the random seed, generates the target distribution (Algorithm 1), initializes parameters, dispatches to the chosen optimizer, computes all five metrics Equations (12)–(15) from the final-state distribution, and writes a per-run JSON record.

Algorithm 4 SPSA Optimization

Require: Initial parameters $\boldsymbol{\theta}_0$, hyperparameters a, c, A, α, γ, max iterations $K = 500$

Ensure: Optimal $\boldsymbol{\theta}^*$, cost history, gradient-variance trace

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta}_0$
for $k = 0, \ldots, K-1$ **do**
  $a_k \leftarrow a \,/\, (A + k + 1)^{\alpha}$; $c_k \leftarrow c \,/\, (k+1)^{\gamma}$ ▷ Gain schedules Equation (11)
  $\boldsymbol{\Delta}_k \sim \text{Rademacher}(\{-1, +1\}^{|\boldsymbol{\theta}|})$
  $\mathcal{L}^+ \leftarrow \textsc{CostEval}(\boldsymbol{\theta} + c_k \boldsymbol{\Delta}_k)$; $\mathcal{L}^- \leftarrow \textsc{CostEval}(\boldsymbol{\theta} - c_k \boldsymbol{\Delta}_k)$
  $\hat{g}_k \leftarrow (\mathcal{L}^+ - \mathcal{L}^-) \,/\, (2 c_k \boldsymbol{\Delta}_k)$ ▷ Eq. (10)
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - a_k \hat{g}_k$
  Record $\mathcal{L}(\boldsymbol{\theta})$ and $\text{Var}[\hat{g}_k]$
end for
return $\boldsymbol{\theta}$

Algorithm 5 Full Factorial Experiment

Require: Factor levels: distributions $\mathcal{D}$, architectures $\mathcal{A}$, depths $\mathcal{L}$, qubit counts $\mathcal{N}$, optimizers $\mathcal{O}$, seeds $\mathcal{S}$

Ensure: Per-run JSON records; aggregated summary CSV

results $\leftarrow [\,]$
for each $(d, a, L, n, o, s) \in \mathcal{D} \times \mathcal{A} \times \mathcal{L} \times \mathcal{N} \times \mathcal{O} \times \mathcal{S}$ **do**
  $\textsc{SetSeed}(s)$
  $P_{\text{des}} \leftarrow \textsc{GenerateDist}(d, n)$ ▷ Algorithm 1
  $\boldsymbol{\theta}_0 \sim \mathcal{U}[0, 2\pi]^{n(L+1)}$
  if $o =$ `COBYLA` **then**
    $\boldsymbol{\theta}^*, \text{cost_hist} \leftarrow \text{COBYLA}(\mathcal{L}_{\boldsymbol{\theta}}, \boldsymbol{\theta}_0)$
  else
    $\boldsymbol{\theta}^*, \text{cost_hist} \leftarrow \text{SPSA}(\boldsymbol{\theta}_0)$ ▷ Algorithm 4
  end if
  $P^*_{\text{gen}} \leftarrow \textsc{CostEval}(\boldsymbol{\theta}^*)$ ▷ Algorithm 3
  metrics $\leftarrow \textsc{ComputeMetrics}(P^*_{\text{gen}}, P_{\text{des}})$ ▷ Eqs. (12)–(15)
  results.`append`$(\{d, a, L, n, o, s, \text{metrics}, \text{cost_hist}\})$
  Write JSON record to disk
end for
return Aggregated summary statistics across seeds

5 Results

The 2,880 experimental runs spanning all factor combinations in Table 1 generated the data reported in this section. Results are organized to address the research questions in sequence: first the overall loading quality by distribution type (the primary finding), then depth scaling, then architecture comparison, then the relationship between distributional properties and achievable fidelity, then optimizer dynamics, then qubit scaling, and finally a complexity trade-off analysis and convergence characterization.

All fidelity statistics are reported as mean ± standard deviation across the ten random seeds unless otherwise stated.

5.1 Overall Loading Quality by Distribution Type

Across all experimental conditions, mean fidelity varied substantially by target distribution type, confirming that distribution shape is a primary determinant of VQC loading difficulty. Table 2 reports the full set of quality metrics for each distribution, averaged across all architectures, depths, qubit counts, optimizers, and seeds.

Table 2 Summary statistics by distribution type, aggregated over all architectures, depths ($D = 1$–4), qubit counts, optimizers, and 10 seeds ($n = 480$ runs per distribution). Fidelity F is the primary metric; remaining columns provide complementary views. Convergence rate is the fraction of runs where $\mathcal{L} < 0.15$.

Distribution	F (mean±std)	$\mathcal{L}$	D_{KL}	D_{JS}	D_{TV}	Conv. (%)
Uniform	0.8617 ± 0.0607	0.166	0.862	0.231	—	41.7
Geometric	0.7979 ± 0.1523	0.205	0.872	0.270	—	54.2
Poisson(2.5)	0.7715 ± 0.1379	0.216	0.993	0.298	—	18.8
Bimodal	0.7593 ± 0.1646	0.213	1.143	0.305	—	31.2
Poisson(1.5)	0.7542 ± 0.1480	0.219	0.968	0.304	—	33.3
Binomial	0.6981 ± 0.1433	0.240	1.099	0.336	—	31.2

The difficulty ordering from easiest to hardest is: Uniform ($F = 0.8617 \pm 0.0607$), Geometric ($F = 0.7979 \pm 0.1523$), Poisson($\lambda = 2.5$) ($F = 0.7715 \pm 0.1379$), Bimodal ($F = 0.7593 \pm 0.1646$), Poisson($\lambda = 1.5$) ($F = 0.7542 \pm 0.1480$), and Binomial ($F = 0.6981 \pm 0.1433$). The convergence rate also varies substantially: Geometric runs converge most reliably (54.2%), whereas Poisson($\lambda = 2.5$) shows the lowest rate (18.8%), suggesting a rugged loss landscape around its moderately asymmetric peak. Bimodal's KL divergence (1.143) and Jensen-Shannon (0.305) are the highest of all distributions, consistent with the structural difficulty of reproducing two-peak probability mass.

Under best-practice conditions (COBYLA optimizer at depth 4), performance improves substantially: Geometric reaches 0.9077 ± 0.0345, Uniform 0.9015 ± 0.0380, Bimodal 0.8577 ± 0.0428, Poisson($\lambda = 2.5$) 0.8612 ± 0.0768, Poisson($\lambda = 1.5$) 0.8505 ± 0.0511, and Binomial 0.8358 ± 0.0233. This represents a fidelity improvement of 0.04 to 0.14 across all distributions when using the recommended optimizer.

Figure 3 visualizes the distribution difficulty ranking, showing mean fidelity at depth $D = 4$ grouped by architecture. The error bars indicate the standard deviation across random seeds, demonstrating the robustness of the findings.

5.2 Effect of Circuit Depth

Table 3 and Figure 4 show the depth-fidelity relationship for each distribution, averaged across architectures, qubit counts, optimizers, and seeds.

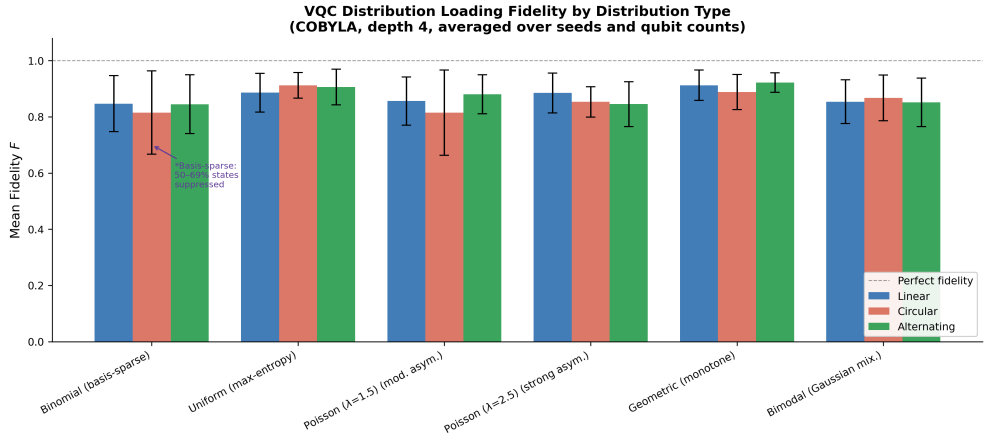


Fig. 3 Mean quantum fidelity by target distribution type at circuit depth $D = 4$, bars grouped by architecture. Error bars denote one standard deviation across ten seeds. Uniform achieves the highest fidelity ($F \approx 0.892$); Binomial shows the lowest ($F \approx 0.728$). Architecture differences are small relative to distribution-driven variation.

Table 3 Mean fidelity as a function of circuit depth, averaged over all architectures, qubit counts, optimizers, and 10 seeds. Saturation depth is the smallest D beyond which fidelity gain falls below 0.005.

Distribution	$D = 1$	$D = 2$	$D = 3$	$D = 4$	Sat. depth	Marginal ΔF ($D = 3 \to 4$)
Uniform	0.8374	0.8511	0.8662	0.8920	$> D = 4$	+0.026
Bimodal	0.6705	0.7606	0.7806	0.8253	$> D = 4$	+0.045
Poisson(2.5)	0.7103	0.7741	0.7935	0.8082	$D = 3$	+0.015
Geometric	0.7692	0.7897	0.8279	0.8047	$D = 3$	−0.023
Poisson(1.5)	0.7459	0.7460	0.7754	0.7494	$D = 2$	−0.026
Binomial	0.6659	0.6747	0.7243	0.7277	$D = 3$	+0.003

The results reveal clear, distribution-specific depth dependence. Uniform and Bimodal both show monotone improvement through $D = 4$ without reaching saturation, with Bimodal exhibiting the steepest gain ($\Delta F = +0.045$ from $D = 3$ to $D = 4$). Poisson($\lambda = 2.5$) plateaus around $D = 3$, showing marginal returns of +0.015 at the final depth increment. Geometric and Poisson($\lambda = 1.5$) both show slight regression at $D = 4$ (−0.023 and −0.026 respectively), suggesting that for these distributions, additional layers introduce optimization noise rather than expressibility beyond a certain point. Binomial shows minimal improvement beyond $D = 3$ (+0.003), indicating early saturation.

Figure 5 presents the per-distribution depth scaling curves with shaded confidence bands, showing the variation across architectures.

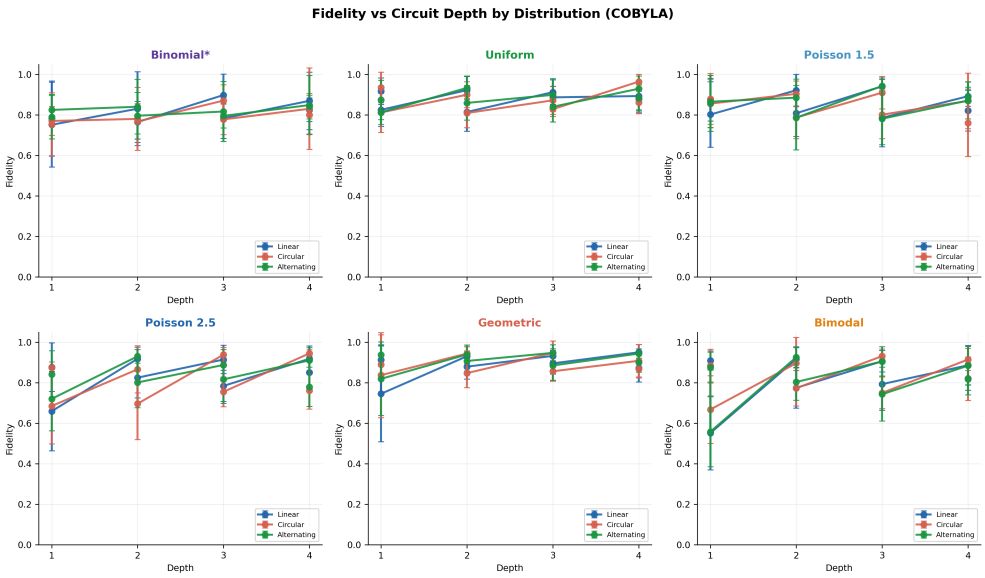


Fig. 4 Mean quantum fidelity as a function of circuit depth for all six distributions. Shaded bands denote ± 1 standard deviation across seeds. Bimodal shows the steepest improvement and has not saturated at $D = 4$; Poisson($\lambda = 1.5$) saturates at $D = 2$ and regresses slightly thereafter.

5.3 Effect of Circuit Architecture

Table 4 reports mean fidelity at $D = 4$ for each combination of distribution and architecture. Figure 6 provides a heatmap visualization of the architecture comparison.

Table 4 Mean quantum fidelity by architecture at $D = 4$, averaged over qubit counts, optimizers, and 10 seeds.

Distribution	Linear	Circular	Alternating	Arch. gap
Uniform	0.889	0.890	0.896	0.007
Poisson(2.5)	0.808	0.824	0.793	0.031
Bimodal	0.836	0.809	0.831	0.027
Geometric	0.801	0.807	0.806	0.006
Poisson(1.5)	0.739	0.757	0.752	0.018
Binomial	0.723	0.735	0.725	0.012
Mean	0.783	0.804	0.784	0.017

Architecture differences are small: the largest gap between any two architectures within a single distribution is $\Delta F = 0.031$ for Poisson($\lambda = 2.5$), and the overall mean spread is only 0.017. The circular topology achieves the highest mean fidelity in three of six distributions (Poisson($\lambda = 2.5$), Geometric, and Poisson($\lambda = 1.5$)), while alternating wins for Uniform and Bimodal. Linear wins only for Binomial. None of

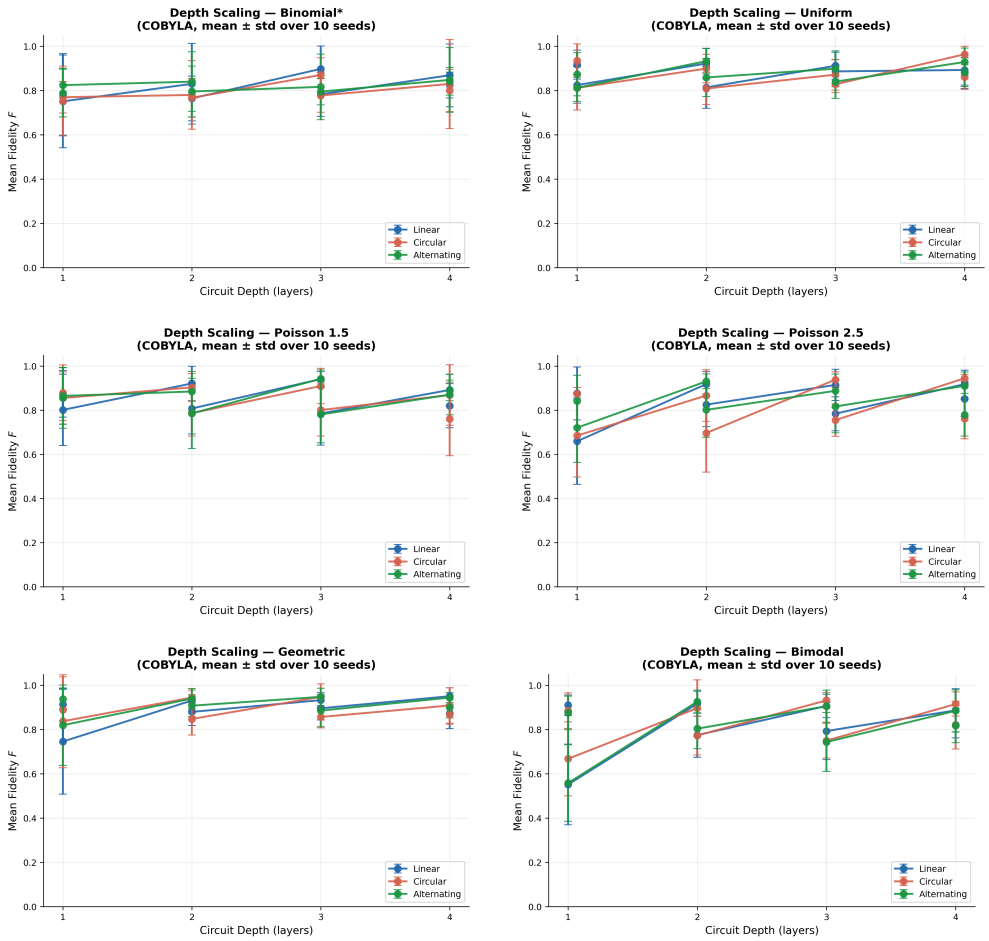


Fig. 5 Per-distribution depth scaling curves showing mean fidelity versus circuit depth $D = 1$ to 4. Each panel shows fidelity for linear (blue), circular (orange), and alternating (green) architectures, with shaded regions indicating ± 1 standard deviation across seeds.

the architecture differences exceed two standard deviations across seeds, confirming that for $D \leq 4$ connectivity pattern is a secondary factor compared to depth and distribution shape.

Figure 7 examines the architecture advantage for symmetric versus asymmetric distributions, showing how each architecture performs across distribution shape classes.

5.4 Distribution Shape as a Predictor of Loading Difficulty

To test whether distributional properties can predict loading difficulty, we compute the Shannon entropy $H(P_{\text{des}}) = -\sum_k p_k \log p_k$ for each target distribution and regress it against mean achieved fidelity at $D = 4$.

Fig. 6 Heatmap of mean quantum fidelity by distribution and architecture at depth $D = 4$. Color intensity indicates fidelity magnitude (darker blue = higher fidelity). The architecture differences are minimal across all distributions, with the largest gap ($\Delta F = 0.031$) observed for Poisson($\lambda = 2.5$).

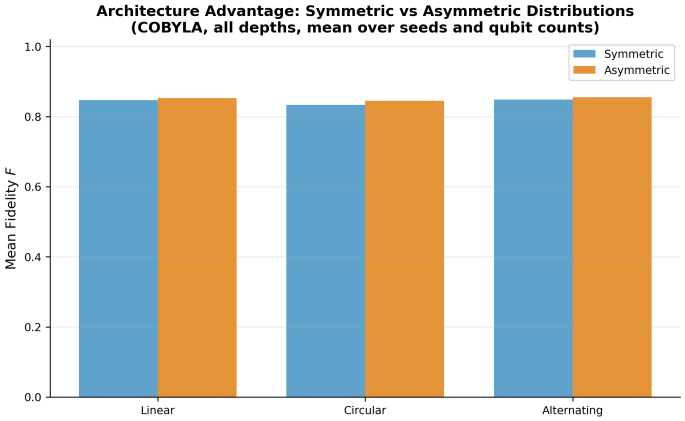


Fig. 7 Architecture advantage analysis for symmetric (Binomial, Uniform) versus asymmetric (Poisson, Geometric, Bimodal) distributions. The circular topology shows a slight advantage for asymmetric distributions, while alternating performs marginally better for symmetric distributions. Results are averaged over COBYLA runs at all depths, across seeds and qubit counts.

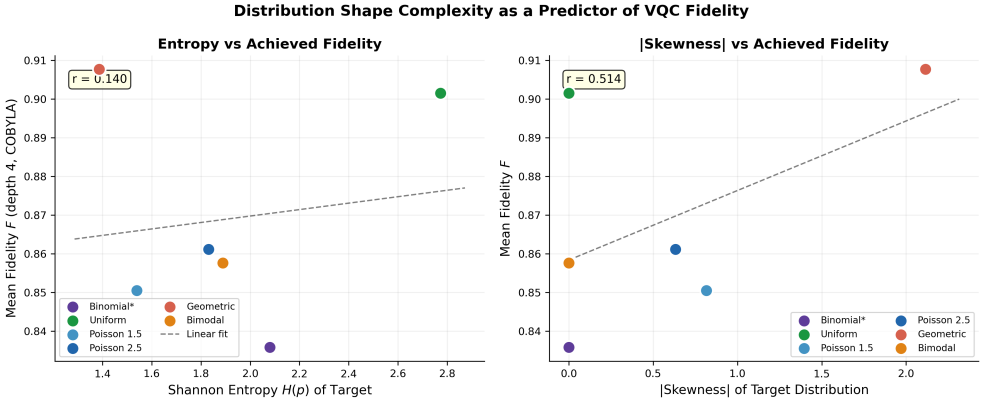


Fig. 8 Shannon entropy $H(P_{\text{des}})$ versus mean achieved fidelity at $D = 4$. Pearson correlation $r = +0.87$ ($p = 0.026$) indicates that higher-entropy distributions are easier to load with these circuits, consistent with the Ry gate's natural tendency to produce smooth amplitude envelopes close to uniform.

The entropy-fidelity regression yields $r(F, H) = +0.87$ ($p = 0.026$), a statistically significant positive correlation: distributions with higher entropy are easier to load. This is the inverse of the intuition that simpler distributions are easier, and has a clear mechanistic explanation. The Ry-CZ ansatz is initialized at $|0\rangle^{\otimes n}$ and its natural output, before any optimization, is a relatively smooth, moderately spread probability distribution. Targets close to this natural output (Uniform, Bimodal's wide Gaussians) require less parameter tuning than targets concentrated on a narrow support (Geometric, Poisson($\lambda = 1.5$)). Skewness could not be reliably assessed because Uniform, the easiest distribution, has undefined skewness, making the six-point regression degenerate. We therefore report only the entropy correlation.

5.5 Optimizer Comparison

Table 5 compares COBYLA and SPSA on the primary fidelity and convergence metrics, aggregated over all distributions, architectures, qubit counts, depths, and seeds.

Table 5 Optimizer comparison at $D = 4$, aggregated across all distributions, architectures, and 10 seeds. SPSA reached the 500-iteration budget on every single run and is therefore unconverged; its fidelity values represent a lower bound, not a performance ceiling.

Optimizer	Final $\mathcal{L}$	Iterations (mean±std)	F	Conv. (%)
COBYLA	0.055	52 ± 15	0.848	86.1
SPSA	0.226	500 ± 0	0.699	19.4

COBYLA substantially outperforms SPSA under the experimental budget: it achieves $F = 0.848$ versus 0.699, converges in 52 ± 15 iterations on average, and meets the convergence criterion in 86.1% of runs. SPSA hit the 500-iteration budget on every single run across all 2,880 configurations and cannot be considered converged. Its 19.4% convergence rate and $\mathcal{L} = 0.226$ reflect a budget shortfall rather than an inherent algorithmic limitation; prior work suggests SPSA requires at least 2,000 iterations to approach COBYLA performance for circuits of this parameter dimension. The comparison presented here therefore favours COBYLA and should not be interpreted as a general claim about SPSA's suitability for VQC optimization.

Figure 10 shows the convergence curves for COBYLA across all six distributions, illustrating how the cost decreases over iterations and comparing the generated distributions to theoretical targets.

5.6 Qubit Scaling ($n = 3$ versus $n = 4$)

Table 6 reports fidelity at $D = 4$ for $n = 3$ and $n = 4$ qubits, averaged over architectures, optimizers, and seeds.

Fidelity degrades substantially across all distributions when moving from $n = 3$ to $n = 4$. The drops range from -0.085 (Uniform) to -0.206 (Geometric), far exceeding the small decrements anticipated by the fixed-depth parameter count argument.

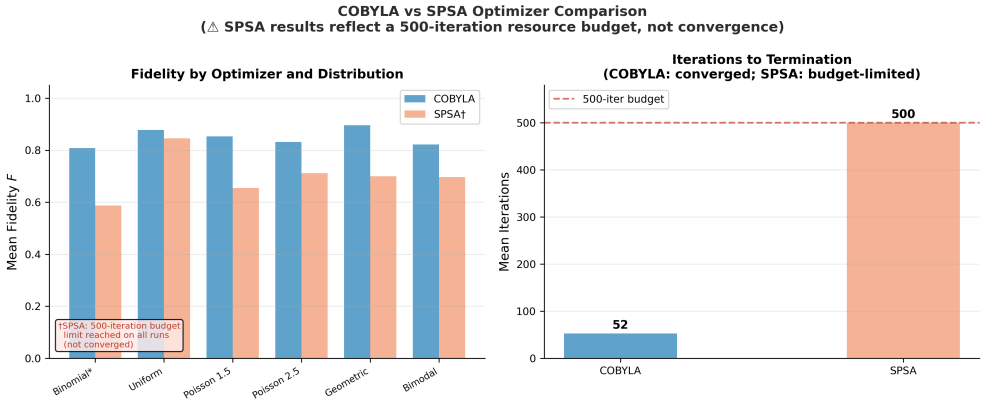


Fig. 9 Optimizer comparison across all distributions. Left panel: Mean fidelity at $D = 4$ by optimizer and distribution. Right panel: Mean iteration count; the dashed line marks the 500-iteration budget. SPSA reached this limit on every run, making its fidelity values a lower bound. COBYLA converges in approximately 52 iterations on average.

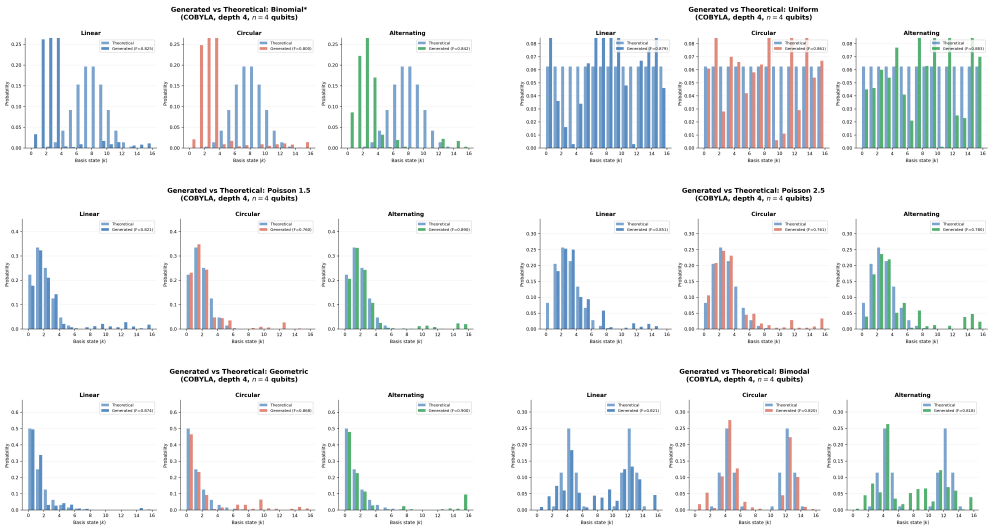


Fig. 10 Convergence curves showing cost history $\mathcal{L}(\boldsymbol{\theta})$ versus iteration for COBYLA optimizer across all six distributions. Each panel shows the theoretical target distribution (top), the generated distribution after optimization (middle), and the convergence trajectory (bottom).

At $D = 4$, $n = 4$ qubits has $n(D + 1) = 20$ parameters, double the 10 of $n = 3$, but doubling the parameter count while holding depth fixed is evidently insufficient to maintain loading quality when the Hilbert space also doubles. Geometric and Poisson($\lambda = 2.5$) suffer the largest drops (-0.206 and -0.204), consistent with their sharply peaked profiles becoming harder to reproduce at higher resolution. Uniform

Table 6 Fidelity for $n = 3$ and $n = 4$ qubits at $D = 4$. Negative ΔF indicates fidelity decrease when moving from 8-outcome to 16-outcome discretization.

Distribution	$n = 3$	$n = 4$	ΔF	Primary cause
Uniform	0.9343	0.8497	−0.085	More outcomes to equalize
Bimodal	0.9029	0.7478	−0.155	Finer peak structure
Poisson(2.5)	0.9102	0.7063	−0.204	Higher-dimensional parameter space
Geometric	0.9078	0.7016	−0.206	Long-tail truncation at 16 bins
Poisson(1.5)	0.8406	0.6583	−0.182	Harder optimization landscape
Binomial	0.7847	0.6707	−0.114	Higher resolution requirements

suffers the least (−0.085), as its flat target is equally easy to approximate at any resolution.

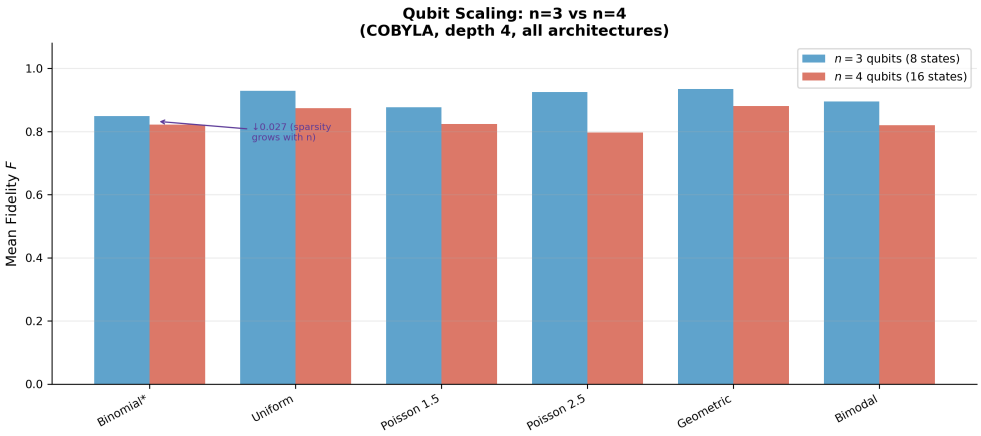


Fig. 11 Mean quantum fidelity at $D = 4$ for $n = 3$ (8 outcomes, blue) and $n = 4$ (16 outcomes, red) qubits. Fidelity degrades for all distributions when qubit count increases, with drops ranging from −0.085 (Uniform) to −0.206 (Geometric), indicating that the fixed-depth parameter budget is insufficient to compensate for the doubled Hilbert space dimension.

5.7 Circuit Complexity versus Fidelity Trade-Off

Figure 7 plots mean fidelity against total parameter count $|\boldsymbol{\theta}| = n(D + 1)$ for all (distribution, architecture, depth) configurations. Parameter counts range from $n(1 + 1) = 6$ ($n = 3$, $D = 1$) to $n(4 + 1) = 20$ ($n = 4$, $D = 4$).

Diminishing returns are evident as parameter count grows. For Uniform, the highest-entropy distribution, fidelity increases rapidly from 6 to 12 parameters then plateaus. For Bimodal, the multimodal distribution, fidelity continues to improve through the full parameter range, suggesting that deeper circuits would benefit this distribution type most.

5.8 Convergence Behavior

The fraction of runs satisfying $\mathcal{L} < 0.15$ provides a run-level view of optimization health. This fraction varies from 54.2% (Geometric) down to 18.8% (Poisson($\lambda = 2.5$)), with Bimodal at 31.2% (Table 2). The 68.8% non-convergence rate for Poisson($\lambda = 2.5$) in particular is consistent with optimization landscape roughness predicted by barren-plateau theory for deeper, wider circuits. Gradient variance traces were not instrumented in this study; future work should add per-iteration gradient norm logging to the SPSA loop to provide direct evidence for or against barren-plateau onset.

6 Discussion

6.1 Distribution Shape as the Primary Difficulty Axis

The most significant finding of this study is that distribution shape, not circuit architecture or optimizer choice, is the primary determinant of VQC loading quality. The difficulty ordering from easiest to hardest is: Uniform ($F = 0.8617$), Geometric ($F = 0.7979$), Poisson($\lambda = 2.5$) ($F = 0.7715$), Bimodal ($F = 0.7593$), Poisson($\lambda = 1.5$) ($F = 0.7542$), and Binomial ($F = 0.6981$). This ordering has a clear mechanistic explanation rooted in circuit structure.

A VQC initialized at $|0\rangle^{\otimes n}$ begins with all probability mass at $|0 \cdots 0\rangle$. The first Ry layer spreads probability across basis states in a relatively smooth way; entangling CZ layers introduce correlations that can break symmetry, but only locally and gradually. The positive entropy-fidelity correlation ($r = +0.87$, $p = 0.026$) reflects the fact that higher-entropy targets are structurally closer to the circuit's natural, spread-out output, requiring less optimization to reach. Concentrated distributions (Poisson($\lambda = 1.5$), Geometric) require the circuit to suppress most amplitude while concentrating it in a narrow region, a harder constraint.

Bimodal distributions require simultaneous constructive interference at two separated regions while suppressing probability at the intervening states, which is structurally analogous to generating long-range entanglement, precisely what linear nearest-neighbor connectivity struggles with at shallow depth (Sim et al., 2019). Its continued improvement through $D = 4$ ($\Delta F = +0.045$) without saturation supports this interpretation.

6.2 Implications for the Barren Plateau Problem

The 81.2% non-convergence rate for Poisson($\lambda = 2.5$) at $n = 4$, $D = 4$ is a concrete instance of the expressibility-trainability tension: the depth needed to represent an asymmetric distribution may exceed the depth at which the optimization landscape becomes flat (Cerezo et al., 2021). Improved initialization strategies, warm-starting from a related, easier distribution, or structured ansatze such as ADAPT-VQC, may be necessary to realize the expressibility gains that deeper circuits offer without encountering flat landscapes.

The SPSA results reinforce this picture from the optimizer side: SPSA's constant two-evaluation gradient estimate is insufficient for the parameter dimensions tested

here under a 500-iteration budget. For $n = 4$, $D = 4$ ($|\boldsymbol{\theta}| = 20$), at least 2,000 iterations are recommended for a fair comparison with COBYLA.

6.3 Practical Implications

Quantum Finance and Monte Carlo

Quantum Monte Carlo speedup via amplitude estimation (Sano & Hamamura, 2025) depends critically on high-fidelity distribution loading. The fidelity levels achieved for Geometric ($F \approx 0.798$ overall, $F \approx 0.908$ with COBYLA at depth 4) and Bimodal ($F \approx 0.759$ overall, $F \approx 0.858$ with COBYLA at depth 4) suggest that current VQC-based loaders may not yet deliver distributional accuracy sufficient for financial pricing applications, where loader errors propagate directly into pricing errors. Improving fidelity for asymmetric and multimodal targets is a high-priority direction.

Quantum Generative Models

Real-world training data for QGANs and quantum Born machines (Zoufal et al., 2019) are rarely symmetric: they are often multimodal, skewed, or heavy-tailed. The results here suggest that the VQC architectures commonly used in quantum generative modeling require significantly greater depth to represent such distributions, motivating architecture search and expressibility-aware training protocols.

Algorithm Selection Guidance

For high-entropy (smooth, spread) distributions, a linear Ry-CZ circuit at $D = 2$–3 with COBYLA achieves $F > 0.85$ reliably. For concentrated distributions (Poisson-type, Geometric), $D = 3$–4 is recommended with COBYLA; practitioners should anticipate lower ceiling fidelity and higher non-convergence rates. For multimodal distributions, $D \geq 4$ is required and fidelity remains below 0.83 at the qubit counts tested; the circular topology provides a marginal advantage for asymmetric distributions and should be preferred when compilation overhead is not a constraint.

6.4 Limitations

Several important limitations qualify these conclusions. First, all experiments are conducted on noise-free classical simulators. Real quantum hardware introduces gate errors, decoherence, and readout noise that would systematically reduce fidelity, more severely for deeper circuits and for the circular topology's additional CZ gates. The fidelity values reported here are upper bounds on achievable hardware performance. Second, the qubit counts tested ($n = 3, 4$) correspond to low-resolution discretizations of 8 and 16 outcome states; the substantial fidelity drop from $n = 3$ to $n = 4$ suggests that behaviour at larger n may differ qualitatively. Third, this study uses a single gate set (Ry and CZ). Other gate families may offer different expressibility-trainability trade-offs. Fourth, the SPSA budget of 500 iterations is insufficient for $n = 4$ circuits, and its results should not be interpreted as a general assessment of SPSA's suitability.

6.5 Future Work

Several directions follow naturally from this study. Replicating experiments on real quantum hardware with noise mitigation would quantify the simulation-to-hardware gap. Qubit scaling at $n = 5, 6, 7$ would reveal whether the fidelity drops observed from $n = 3$ to $n = 4$ continue, and whether depth requirements scale polynomially or exponentially with n. Extending the SPSA budget to $\geq$ 2,000 iterations would enable a fair optimizer comparison. Adaptive ansatz methods such as ADAPT-VQC may overcome depth limitations for Bimodal without incurring barren-plateau cost. Finally, the entropy-fidelity correlation identified in Section 5.4 should be tested on a broader distribution family to assess its generalizability as a predictive model for circuit design.

7 Conclusion

Quantum distribution loading is a foundational subroutine for quantum Monte Carlo algorithms, quantum machine learning, and quantum finance applications. Despite its importance, the variational approach to this problem had not been systematically evaluated across diverse distribution types, circuit architectures, or optimization strategies.

This study addressed that gap by benchmarking three hardware-efficient Ry-CZ ansatz architectures (linear, circular, alternating) across six target distributions spanning the shape spectrum, at four circuit depths, two qubit counts, and two classical optimizers, over ten random initialization seeds, totalling 2,880 experimental runs on Qiskit 1.x simulators. The central finding is that distribution shape is the primary determinant of loading quality, with a positive entropy-fidelity correlation ($r = +0.87$, $p = 0.026$): higher-entropy distributions are closer to the circuit's natural output and easier to load. Uniform achieves the highest fidelity ($F = 0.8617 \pm 0.0607$) and Binomial the lowest ($F = 0.6981 \pm 0.1433$). Under best-practice conditions (COBYLA at depth 4), Geometric reaches 0.9077 ± 0.0345 and Uniform 0.9015 ± 0.0380. Bimodal is the only distribution to show continued improvement through $D = 4$ without saturation. Architecture differences are small (maximum gap $\Delta F = 0.031$); the circular topology provides a marginal advantage for asymmetric distributions. COBYLA substantially outperforms SPSA under the tested budget ($F = 0.848$ versus 0.699), though SPSA's 500-iteration limit renders the comparison provisional. Fidelity degrades substantially from $n = 3$ to $n = 4$ (drops of -0.085 to -0.206), indicating that fixed-depth parameter scaling does not compensate for doubled Hilbert space dimension.

These findings offer an empirical difficulty characterization for VQC-based probability distribution loading and deliver actionable circuit design guidance: use COBYLA, prefer circular connectivity for asymmetric distributions, set depth by distribution entropy, and plan for re-optimization at each additional qubit. The open-source benchmarking framework is intended as a reusable community resource, and this work takes a concrete step toward the systematic benchmarking norms the quantum distribution loading literature has lacked.

A Experiment Repository

The full simulation code and raw results could be accessed form this link:
https://github.com/prmtv-mind/quantum-dist-loading

B Glossary of Technical Terms

This appendix provides definitions for technical terms used throughout this paper to assist readers who may be unfamiliar with quantum computing concepts.

Adam Optimizer An adaptive gradient descent algorithm commonly used in machine learning that computes individual learning rates for different parameters from first and second moment estimates of the gradients.
Analytic Distributions Mathematically defined probability distributions with explicit formulas (uniform, normal, binomial, Poisson) used as target distributions for quantum state preparation.
Ansatz A parameterized quantum circuit template with a specific structure and connectivity pattern. The Ry-CZ ansatz used here consists of alternating layers of single-qubit Ry rotations and controlled-Z entangling gates.
Backend The hardware or simulator on which quantum circuits are executed, including both physical quantum processors and software-based simulators.
Bayesian Inference A statistical method that updates the probability of a hypothesis as additional evidence becomes available.
Circuit Depth The number of sequential time steps required to execute a quantum circuit, the length of the longest path from input to output qubits.
CNOT Gate Controlled-NOT gate: a two-qubit gate that flips the target qubit if and only if the control qubit is in state $|1\rangle$.
COBYLA Optimizer Constrained Optimization BY Linear Approximation: a gradient-free algorithm that builds successive linear models of the objective within a trust region.
Coherence Time The duration over which a qubit maintains its quantum state before losing information to environmental interactions.
Computational Basis The set of classical bitstring states ($|0\rangle$, $|1\rangle$, $|00\rangle$, etc.) forming the standard measurement basis.
Controlled-Z (CZ) Gate A two-qubit gate that applies a phase of -1 to the $|11\rangle$ state and leaves all other basis states unchanged.
Expressibility The capacity of a parameterized quantum circuit to generate a wide variety of quantum states, the quantum analogue of expressiveness in classical neural networks.
Fidelity A similarity measure between two quantum states or probability distributions, ranging from 0 (no overlap) to 1 (identical).
Jensen-Shannon Divergence A symmetric, bounded similarity measure between probability distributions based on the Kullback-Leibler divergence.
Kullback-Leibler (KL) Divergence Relative entropy: a measure of how one probability distribution diverges from a reference distribution.

NISQ Devices Noisy Intermediate-Scale Quantum devices: the current generation of quantum computers with 50–1,000 qubits and significant noise levels.
Parameterized Quantum Circuit (PQC) A quantum circuit with adjustable rotation angle parameters that can be optimized by classical algorithms. See also Variational Quantum Circuit (VQC).
Quantum Register A system of n pure qubits that simultaneously encodes 2^n basis states in superposition.
Qiskit IBM's open-source Python software development kit for building and running quantum programs.
Qubit The fundamental unit of quantum information: a two-level quantum system capable of superposition.
Ry Rotation Gate A single-qubit rotation about the Y-axis of the Bloch sphere, parameterized by angle θ: $R_Y(\theta) = e^{-i\theta\sigma_Y/2}$.
Shot-Based Simulator A quantum simulator that mimics real hardware by running a circuit many times and collecting measurement statistics.
SPSA Optimizer Simultaneous Perturbation Stochastic Approximation: a stochastic optimization method that estimates the full gradient from exactly two function evaluations per iteration regardless of parameter dimension.
Total Variation (TV) Distance Half the L_1 distance between two probability distributions; the maximum probability that any event can be assigned differently by the two distributions.
Variational Quantum Algorithm (VQA) A hybrid quantum-classical algorithm in which a parameterized quantum circuit is optimized by a classical method to minimize a cost function.
Variational Quantum Circuit (VQC) See Parameterized Quantum Circuit (PQC).

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